Form B

Core Focus on Proportions & Probability

Block 2 Review ~ Proportions and Similarity

ANSWER KEY

Name _____ Period _____ Date _____

Part I – Selected Response

1. What is the value of x in the proportion $\frac{8}{10} = \frac{12}{x}$?

A. 14
B. 15
C. 16
D. 18

B ($8x = 10 \cdot 12 = 120$, so $x = 15$)

2. Lena bought 4 notebooks for $9.80. How much would 7 notebooks cost?

A. $12.95
B. $15.68
C. $17.15
D. $19.60

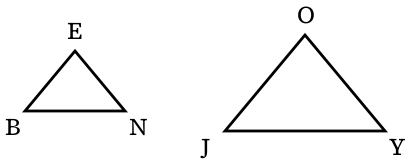
C ($9.80 ÷ 4 = $2.45 each; 7 × $2.45 = $17.15)

3. Marcus biked 4 miles in 18 minutes. How long would it take him to bike 10 miles at this pace?

A. 40 minutes
B. 45 minutes
C. $2\frac{2}{9}$ minutes
D. 72 minutes

B (4/18 = 10/x → 4x = 180 → x = 45 minutes)

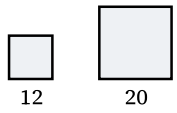
4. △BEN is similar to △JOY. Which side of △JOY corresponds to $\overline{BN}$ in △BEN?



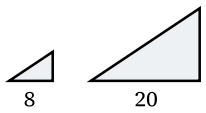
A. $\overline{JO}$
B. $\overline{OY}$
C. $\overline{YO}$
D. $\overline{JY}$

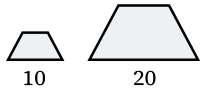
D (B ↔ J and N ↔ Y, so BN ↔ JY)

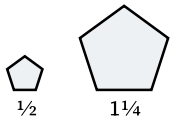
5. Which pairs of similar figures have a scale factor of 2 : 5? Circle all that apply.

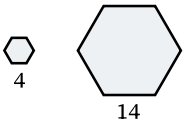
A.


B.


C.


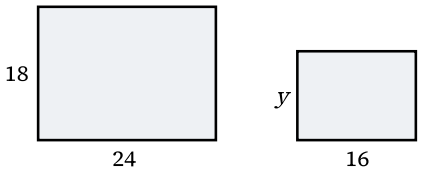
D.


E.


F.


B, C, E (6 : 15, 8 : 20 and ½ : 1¼ all simplify to 2 : 5; 12 : 20 = 3 : 5, 10 : 20 = 1 : 2, 4 : 14 = 2 : 7)

6. What is the value of y in the similar figures?



A. 10
B. 12
C. 14
D. 21

B (24/16 = 18/y → 24y = 288 → y = 12)

7. Two similar rectangles have a scale factor of 3 : 5. For numbers 7a – 7d, determine whether each statement about the rectangles is true or false.

7a. The ratio of their perimeters is 9 : 25. **TRUE FALSE**

7b. The ratio of their areas is 9 : 25. **TRUE FALSE**

7c. Their perimeters could be 24 cm and 40 cm. **TRUE FALSE**

7d. Two corresponding sides could be 6 ft and 8 ft. **TRUE FALSE**

7a FALSE (perimeters are 3 : 5) · 7b TRUE ($3^2 : 5^2$) · 7c TRUE (24 : 40 = 3 : 5) · 7d FALSE (6 : 8 = 3 : 4)

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Part II – Constructed Response

Solve each proportion.

1. $\frac{5}{8} = \frac{20}{x}$

x = 32 (5x = 8 · 20 = 160; check 20/32 = 5/8)

2. $\frac{12}{18} = \frac{y}{45}$

y = 30 (18y = 12 · 45 = 540; check 30/45 = 2/3 = 12/18)

3. Priya bought 3 bags of apples for $7.47. How much would 5 bags of apples cost? Show all work necessary to justify your answer.

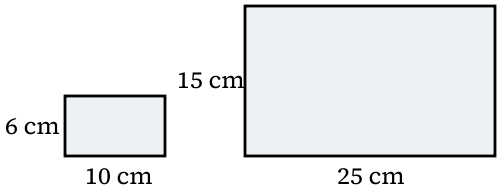
$12.45 (3/7.47 = 5/x → 3x = 37.35 → x = 12.45; or $2.49 per bag × 5 = $12.45)

4. Carlos typed 120 words in 3 minutes. Dana typed 210 words in 5 minutes. Which person typed at a faster rate? Use mathematics to justify your answer.

Dana. Carlos: 120 ÷ 3 = 40 words per minute. Dana: 210 ÷ 5 = 42 words per minute. 42 > 40.

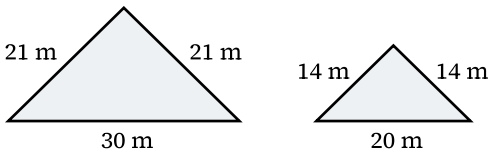
Determine the scale factor for each pair of similar figures.

5.



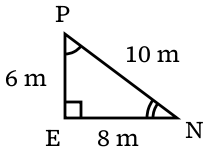
2 : 5 (10 : 25 = 6 : 15 = 2 : 5)

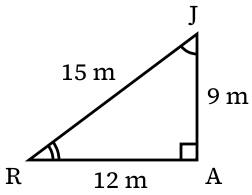
6.



3 : 2 (large to small: 21 : 14 = 30 : 20 = 3 : 2; small to large is 2 : 3)

7. Use the figures below to find the corresponding sides and corresponding angles to the ones identified. Matching arcs show congruent angles.





$\overline{PE}$ corresponds to _____ ∠P ≅ ∠_____

$\overline{EN}$ corresponds to _____ ∠E ≅ ∠_____

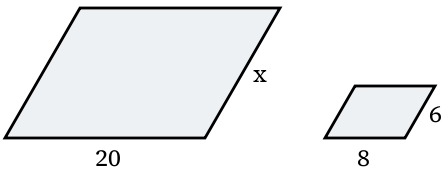
$\overline{PN}$ corresponds to _____ ∠N ≅ ∠_____

Are △PEN and △JAR congruent, similar, or neither? Explain your reasoning.

PE ↔ JA, EN ↔ AR, PN ↔ JR; ∠P ≅ ∠J, ∠E ≅ ∠A, ∠N ≅ ∠R. Similar: the corresponding angles are congruent and the sides are proportional (6/9 = 8/12 = 10/15 = 2/3), but the sides are not equal, so they are not congruent.

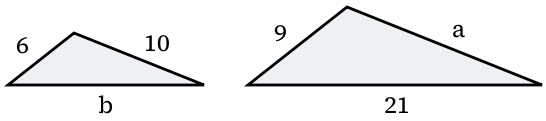
Each pair of shapes below are similar. Solve for each variable. Show all work necessary to justify your answer.

8.



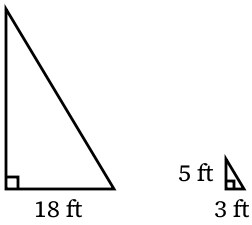
x = 15 (20/8 = x/6 → 8x = 120 → x = 15)

9.



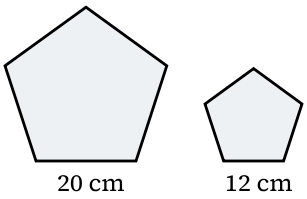
Scale factor 6 : 9 = 2 : 3. a: 6/9 = 10/a → 6a = 90 → a = 15. b: 6/9 = b/21 → 9b = 126 → b = 14.

10. Nora wants to know the height of the flagpole at her school. One afternoon, the flagpole's shadow along the ground is 18 feet long. At the same time, Nora's own shadow is 3 feet long. Nora is 5 feet tall. Find the height of the flagpole. Use mathematics to justify your answer.



30 feet (h/5 = 18/3 → 3h = 90 → h = 30; the flagpole's shadow is 6 times Nora's, so it is 6 × 5 = 30 ft tall)

11. Use the similar regular pentagons below.



a. Find their scale factor (large to small).

5 : 3 (20 : 12)

b. Find the ratio of their perimeters.

5 : 3

c. Find the ratio of their areas.

25 : 9 ($5^2 : 3^2$)

d. Suppose the larger pentagon has a perimeter of 100 cm. What is the perimeter of the smaller pentagon? Show all work necessary to justify your answer.

60 cm (5/3 = 100/x → 5x = 300 → x = 60; check 5 × 12 = 60)

12. Two similar rectangles have perimeters of 12 feet and 48 feet. The smaller rectangle has an area of 5 ft^2. Marco used this information to find the area of the larger rectangle. Unfortunately, he made an error in his work. Explain the error and then find the correct area of the larger rectangle.

> Scale factor is 12 : 48, which is 1 : 4
>
> Ratio of their areas is 1 : 4
>
> Let x = the area of the larger rectangle
>
> $\frac{1}{4} = \frac{5}{x}$
>
> $x = 20$
>
> The area of the larger rectangle is 20 ft^2.

Marco used the scale factor as the ratio of the areas. The area ratio is the scale factor squared: $1^2 : 4^2 = 1 : 16$. 1/16 = 5/x → x = 80. The larger area is 80 ft^2.

13. A map has a scale of 3 centimeters : 40 kilometers. On the map, the distance from Owen's town to the lake is 7.5 centimeters. How many kilometers is the lake from Owen's town? Show all work necessary to justify your answer.

100 km (3/40 = 7.5/x → 3x = 300 → x = 100)

Core Focus on Proportions & Probability

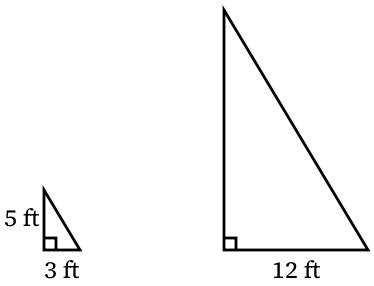
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Part III – Problem Solving

Jada stands next to a light pole in a parking lot. She is 5 feet tall and her shadow along the ground has a length of 3 feet. At the same time, the light pole casts a shadow along the ground that has a length of 12 feet. What would the length of the light pole's shadow along the ground be if the pole were 5 feet shorter?



Pole now: h/5 = 12/3 → 3h = 60 → h = 20 ft. Five feet shorter: 20 − 5 = 15 ft. New shadow: 15/5 = s/3 → 5s = 45 → s = 9 ft. The shadow would be 9 feet long.